

TOOLS For ENGINEERS, CODERS And MEDIA ENTHUSIASTS

The following software, besides other fun stuff, can be found in the DVD accompanying the EFY Plus edition of this magazine

PICSimLab

PICSimLab is an electronics development board emulator. It comes with an integrated MPLABX debugger. It supports PICs and simavr microcontrollers. This open source emulator supports real-time emulation and has support for Arduino board programming.

KDevelop

This free, open source integrated development environment (IDE) is available for Microsoft Windows and other Unix-like operating systems (OSes). The latest version has parser backends for programming languages like C, C++, Javascript/QML, Objective-C and OpenCL. With external plugins, it can support PHP, Ruby and Python 3. The interface is interactive with features like editing, navigation, syntax highlighting, code folding and debugging.

Atom

Atom is an easy-to-use text editor and source code editor compatible with various platforms like Windows, Linux and MacOS. Using plugins, it can act as an IDE for various languages like C/C++, C#, Go, HTML, JAVA, JavaScript and SQL, JSON, Python, Ruby, XML, SCALA, CSS, CoffeeScript and many more. It is highly customisable.

VirtualBox

VirtualBox is an open source virtual machine monitor for x86 and AMD64/Intel 64 computers, created by Innotek GmbH and currently being maintained by Oracle Corp. It can run on a range of OSes including Windows, Linux, MacOS, Solaris and OpenSolaris, and gives port-based support for FreeBSD and Genode. It can create virtualisations for running version and deviations of platforms like Windows, Linux, BSD, OS/2, OS86, Haiku and Solaris, but has

limited support for MacOS virtualisation. VirtualBox can run multiple virtualisations of different OSes simultaneously, and can do operations on each of these independent of others.

CodeLite

This is a free open source IDE for C, C++, PHP and Javascript programs. It is compatible with multiple platforms including Windows, Linux, MacOS X and FreeBSD. The latest version of CodeLite comes with UnitTest++ as its debugger and supports operations like code completion, code refactoring, source browsing and syntax highlighting.

FFmpeg

FFmpeg free project is used for maintaining, editing and managing media files (mainly audio and video). Main components of FFmpeg are three command line tools and its extensive range of libraries. Among command line tools, FFmpeg is used for major file editing operations, video codec updates, real-time encoding of audio and video from various software and hardware sources. FFplay is a media player. FFprobe displays various media file information.

Paint.NET

This Windows-based photo-and-image-editing software's interface is highly user-friendly. Its latest feature is a multi-tabbed interface, denoted by live thumbnails, to work on multiple images simultaneously and toggle between these easily. The software brings a layered image-editing approach, good number of special effects and variety of tools.

GIMP

This free, open source raster graphics editor is compatible across plat-

forms including Linux, Windows and MacOS. The latest version, 2.10.0, adds a number of upgrades. Most of the image processing operations have been ported to Generic Graphics Library. Digital Painting feature now supports canvas rotation and flipping, MyPaintBrush support and symmetry painting, among others. New tools have been added and existing tools have been improved, such as warp transform, unified transform and handle transform.

Krita

Krita is a free, open source image developer and editing tool. Compatible with Windows, MacOS and Linux, Krita comes with an interactive, high-quality canvas accelerated by Open Graphics Library. The latest version utilises Qt5 and KDE Frameworks 5. It comes with a gamut of features like adjustable paint brush tools powered by nine different brush engines, pen stabiliser, multibrush support, settings based on paint commands and much more.

Audacity

Audacity is an audio editing tool, available for free as an open source entity. Supported across almost all major OSes, it brings huge flexibility to modify, arrange and convert audio files. Features include MIDI playback, scrubbing, amplitude editing, audio recording, spectrum analysis, noise reduction, and conversion of audio files from cassettes and tapes to digital files, among others. It is highly customisable and supports various plugins. Support file input and output types include, but are not limited to, MP3, WAV, AIFF, FLAC, WMA and AC3.

Alongside these, there are more useful software that are available in the DVD. **EFY**